

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 03

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Recap of Previous Lecture



Topic

Cache Write

Topic

Write Through & Write Back

Topic

Write Allocate & No Write Allocate

Topics to be Covered



Topic

Cache Mapping

Topic

Direct Mapping

Topic

Tag

Topic

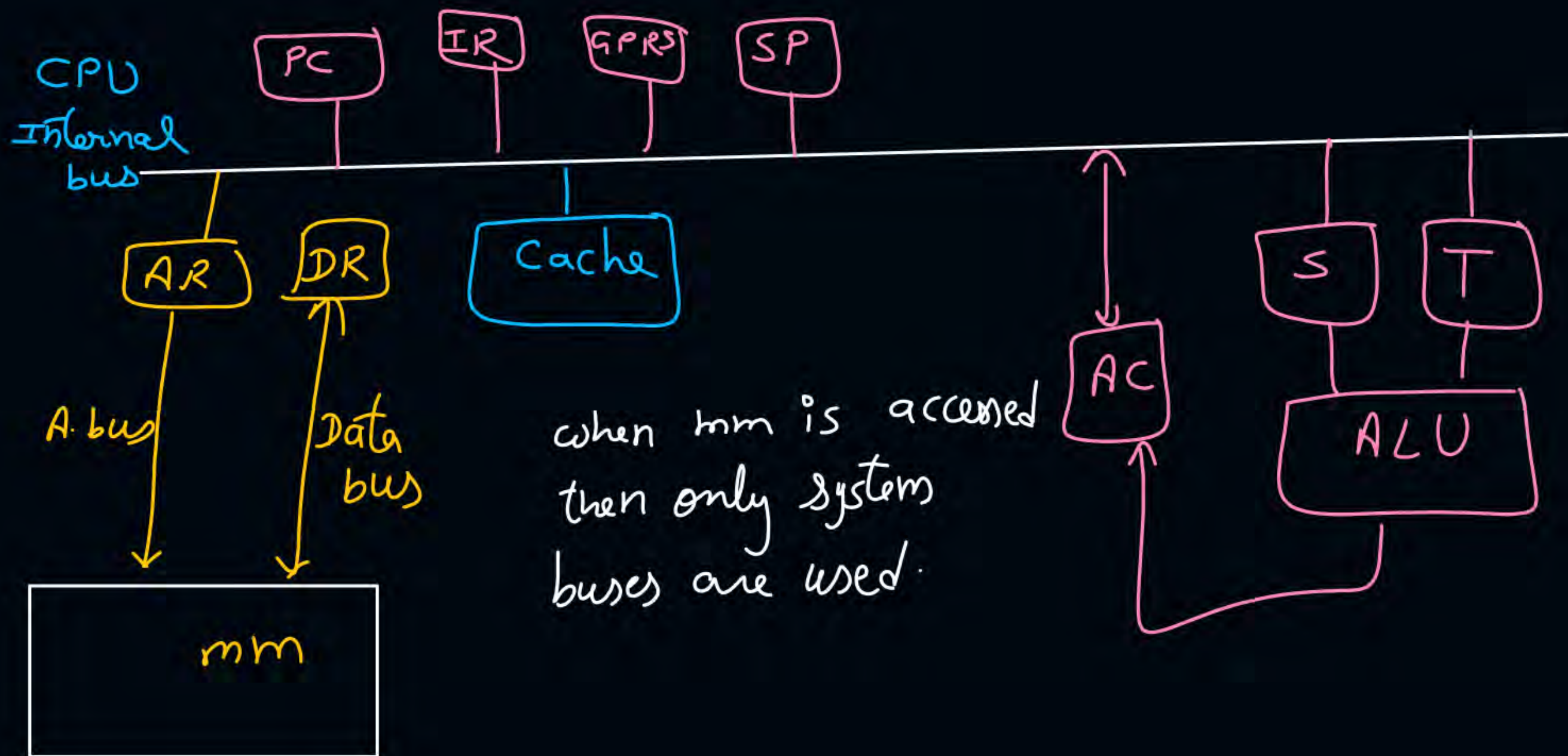
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#Q. Size of data sent to main memory from CPU:

1. For write hit, when a write through cache is used? $\Rightarrow$ 1 byte or 1 data item size
2. For write miss, when a write through cache is used? $\Rightarrow$ 1 byte or 1 data item size
3. For write hit, when a write back cache is used? $\Rightarrow$ 0
4. For write miss, when a write back cache is used? $\Rightarrow$ 0

#Q. Size of data sent from main memory to cache:

1. For write hit, when a write through cache is used? $\Rightarrow 0$
2. For write miss, when a write through cache is used? $\Rightarrow 0$
3. For write hit, when a write back cache is used? $\Rightarrow 0$
4. For write miss, when a write back cache is used? $\Rightarrow 1 \text{ block}$



[Question]

- #Q. Consider a computer with the following features:
- 90% of all memory accesses are found in the cache (hit ratio = 0.9)
 - The block size is 2 words and the whole block is read on any miss
 - The CPU sends references to the cache at the rate of 10^7 words per second
 - 25% of the above references are writes (writes = 25%, reads = 75%)
 - The bus can support 10^7 words per second, read or writes (total bus bandwidth = 10^7)
 - The bus reads or writes a single word at a time
 - Assume at any one time, 30% of the block frames in the cache have been modified

Calculate the percentage of the bus bandwidth used on the average when:

1. Cache is write through with no write allocate
2. Cache is write back with write allocate

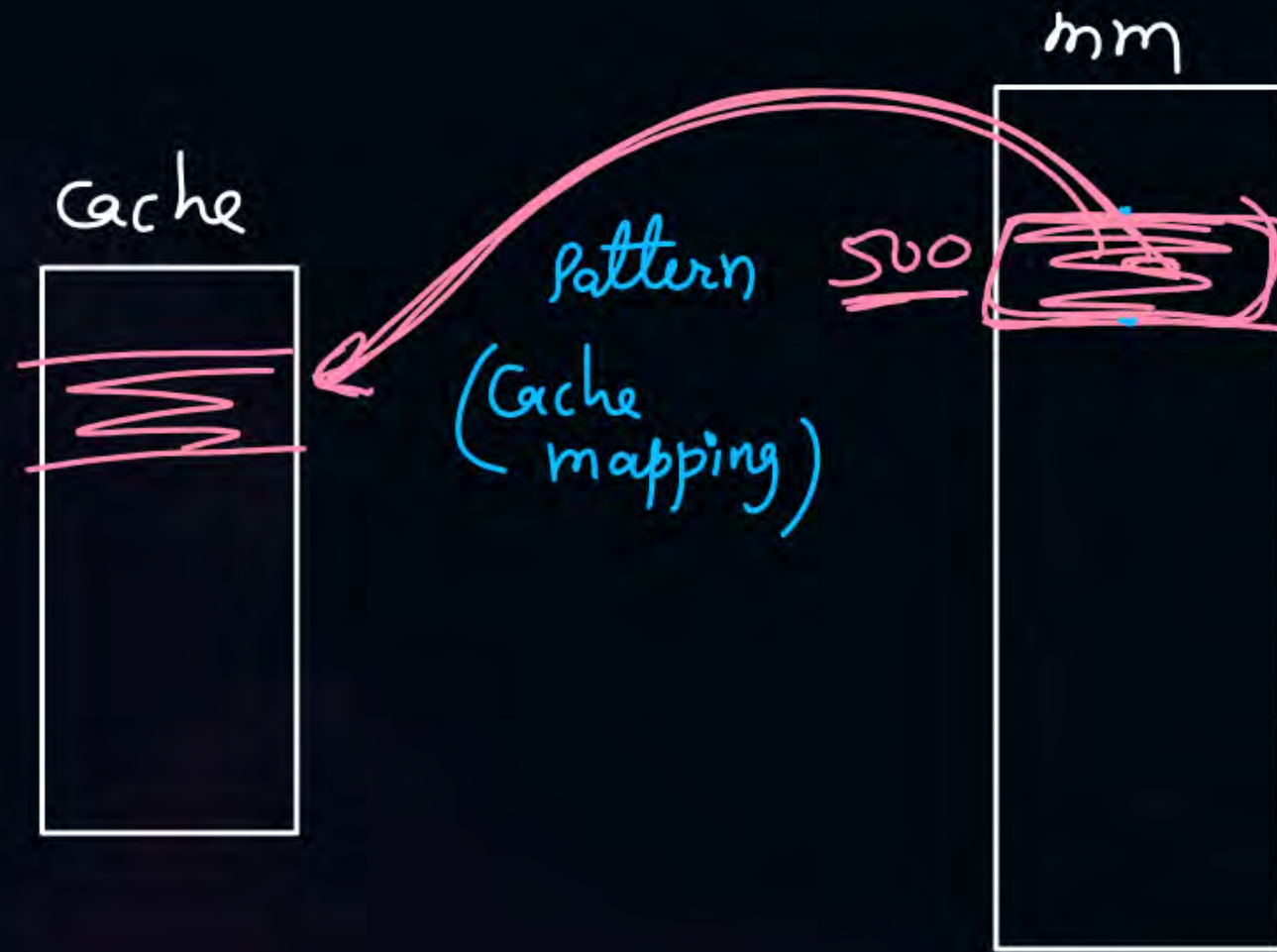


Topic : Cache Mapping

CPU always generates mm address.

$$cm\ local^n = f(mm\ location)$$

mapping is done
on blocks.





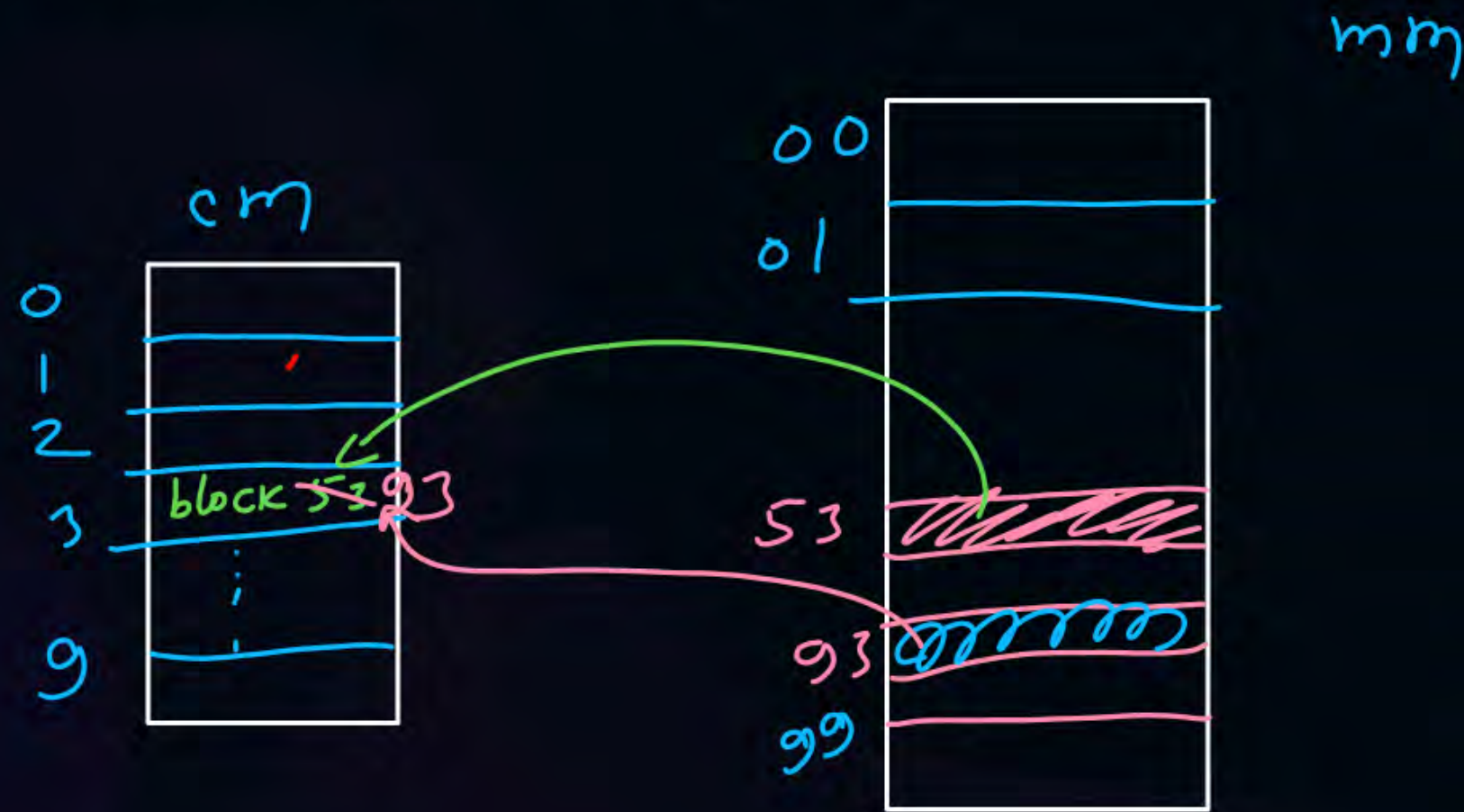
Topic : Cache Mapping

- Direct Mapping
- Set Associative Mapping
- Fully Associative Mapping



Topic : Direct Mapping

- Blocks in cache = 10 (0-9)
- Blocks in Main memory = 100 (00-99)





Topic : Direct Mapping



93
~~33~~

Cache

0 1 2 3 4 5 6 7 8 9

Main
Memory

00	01	02	03	04	05	06	07	08	09
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39
:	:	:	:	:	:	:	:	:	:
90	91	92	93	94	95	96	97	98	99



Topic : Direct Mapping



$$\text{cm block no.} = (\text{mm block no.}) \% (\text{no. of blocks in cache})$$

$$\text{Tag} = \left\lfloor \frac{\text{mm block no.}}{\text{no. of blocks in cache}} \right\rfloor$$



Topic : Direct Mapping

CPU Request (MM block)	Mapping (CM block no.)	Hit /Miss	Comments
53	$53 \% 10 = 3$ goto block 3 in cache & check	Miss	Bring mm block 53 content in cache at block no. 3.
93	$93 \% 10 = 3$ goto block 3 in cache & check ↓ block 53 content is in cache	Miss	bring mm block 93 content in cache at block no 3, by replacing block 53 content.



Topic : Direct Mapping

	Tag	cm blocks
0		
1		
2		
3	59	block 53 93
...	...	...
9		

mm block no.

Tag	cm block no.
-----	--------------

ex:-
mm block no. 53

5	3
Tag	

CPU request mm block 93

9	3
---	---

miss

cm block no. 3

Tag Bits: A small portion of the address used to identify which block of main memory is stored in a particular cache line.

Tag :- It identifies among all competitors which one is present in cache at this block.

one Tag is stored in cache for each block in cache.

no. of tags = no. of blocks in cache



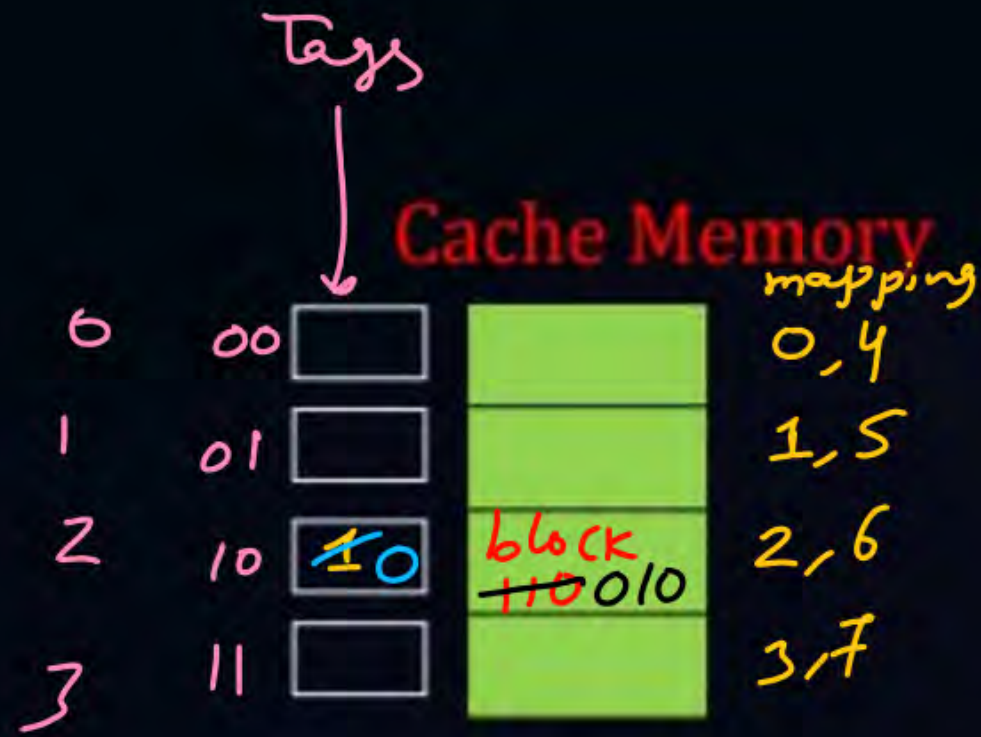
Topic : Direct Mapping



- Blocks in cache = 4 (00-11) 0 to 3
- Blocks in Main memory = 8 (000-111) 0 to 7



Topic : Direct Mapping



Main Memory

000		0
001		1
010		2
011		3
100		4
101		5
110		6
111		7

mm block no.
← 3 bits →

Tag	cm block no.
1 bit	2 bits

no. of blocks in cache = 4
 $= 2^2$



Topic : Direct Mapping

CPU Request (MM block)	Mapping (CM block no.)	Hit /Miss	Comments
$(6)_{10}$ $= (110)_2$	1 10 ↓ ↓ Tag cm block no.	miss	bring mm block 110 content in cache at block 10 with tag <u>1</u> .
$(2)_{10}$ $= (010)_2$	0 10 ↓ ↓ Tag cm block no.	miss	bring mm block 010 content in cache at block 10 by replacing mm block 110 and change tag to 0.



Topic : Direct Mapping

- Blocks in cache = 4 (00-11)
- Blocks in Main memory = 8 (000-111)
- Block Size = 2 Bytes = $2^1 B$
- Size of Cache memory = $4 * 2 = 8 \text{ bytes}$
- Size of Main memory = $8 * 2 = 16 \text{ bytes} = 2^4 B$
- Size of Main memory address = 4 bits
byte $\downarrow$ addressable



Topic : Direct Mapping

cm block no. (line no)
or Index

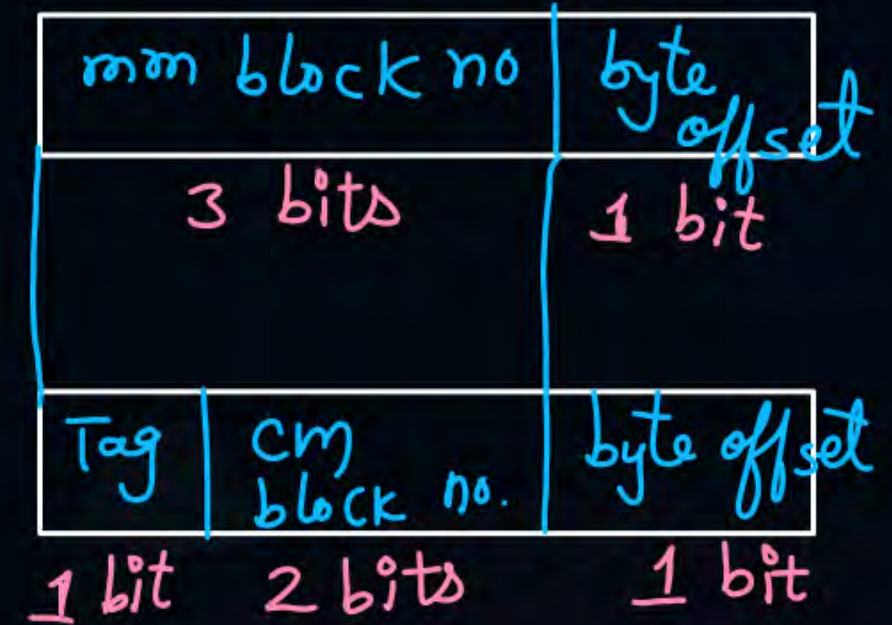


Byte Offset: Specifies the exact byte within the block.

mm addresses



mm add.
4 bits





Topic : Direct Mapping

CPU Request (MM add.)	Mapping(CM block no.)	Hit/Miss	Comments
$(1100)_2$	110 0 mm block no. → byte Tag cm block 1 10 0	miss	bring mm block 110 content in cache at block 10 with tag 1
$(1101)_2$	110 1 1 10 1 Tag cm block no. byte	Hit	CPU accesses byte 1 of this block from Cache.



Topic : Direct Mapping



$$\text{no. of bits in byte offset} = \log_2(\text{block size})$$

$$\text{no. of bits in cm block no.} = \log_2(\text{no. of blocks in cache})$$

$$\text{no. of blocks in cache} = \frac{\text{Cache size}}{\text{block size}}$$

$$\begin{array}{l} \text{Tag directory size} \\ \text{or metadata} \end{array} = \text{no. of blocks in cache} * \text{Tag bits}$$

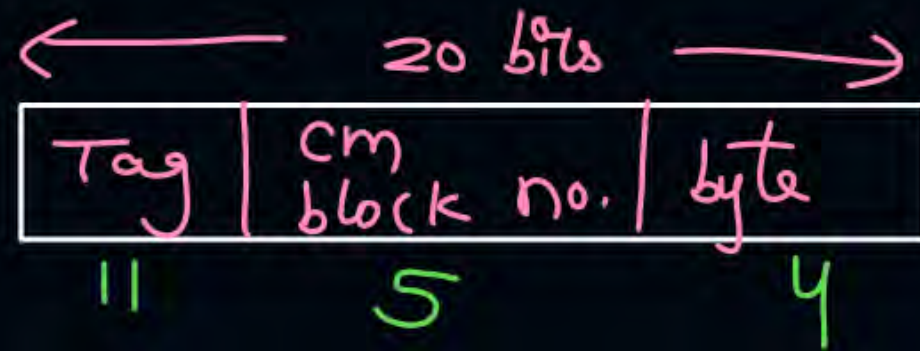
Ques) mm add. = 20 bits

cm size = 512 bytes

block size = 16 bytes = 2^4 B

Direct mapping

↓
byte offset = 4 bits



Tag directory size = $(2^5 * 11)$ bits

Index = 5 bits

Tag = 11 bits

no. of blocks
in cache = $\frac{512 \text{ B}}{16 \text{ B}}$

$$= \frac{2^9}{2^4}$$

$$= 2^5$$

↓

cm block no = 5 bits

Ques) mm add. = 32 bits

cm size = 32 KB

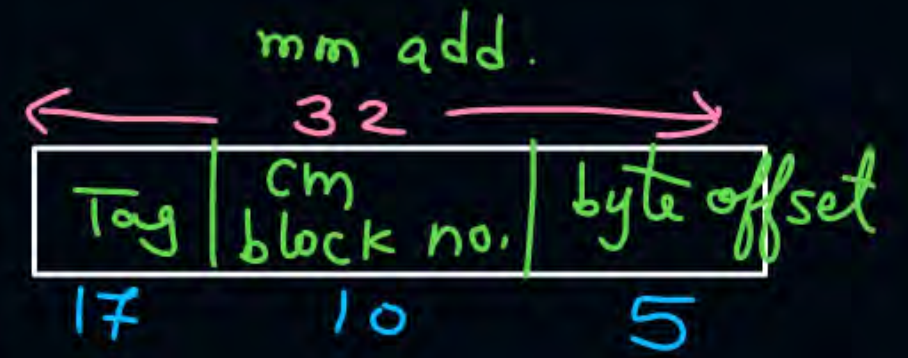
block size = 32 B = 2^5 B = byte offset = 5 bits

direct mapping

Index = $\frac{10}{}$ bits

Tag = $\frac{17}{}$ bits

Tag directory size = $2^{10} * 17$ bits
= 17 k bits



no. of blocks in cache = $\frac{32 \text{ KB}}{32 \text{ B}}$

= 1K

= 2^{10}

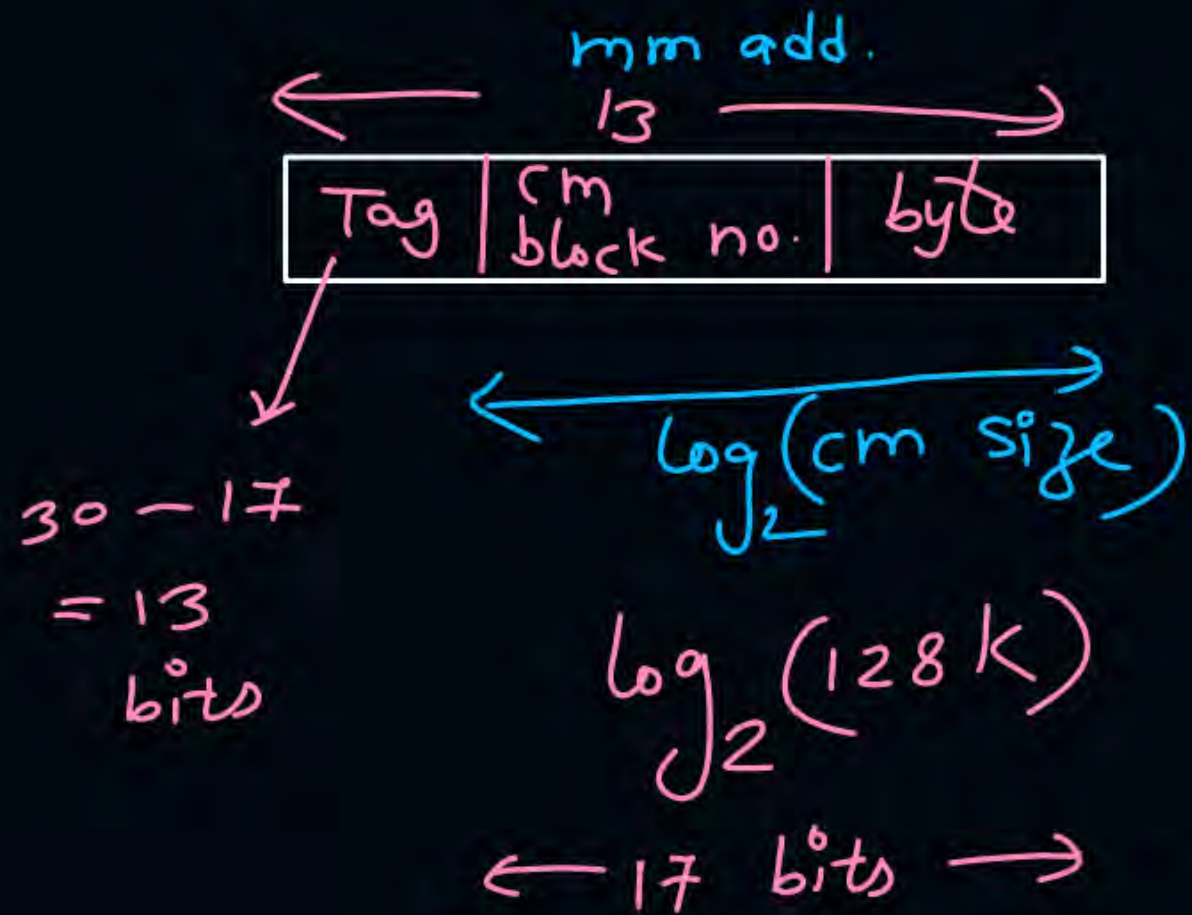
↓

cm block no. = 10

Ques) mm add. = 30 bits

cm size = 128 KB

$$\text{Tag} = \underline{13} \text{ bits}$$



Ques) Direct mapping
cm size = 256 K bytes

Tag = 11 bits

mm size = 0.5 GB ?
↓
byte addressable

mm add.

Tag	block	byte
-----	-------	------

11



$\log_2 256K$

= 18 bits

29 bits

mm size = 2^{29} bytes

= 512 MB

= 0.5 GB

Cache Initialization :-

valid Invalid	Tag	cm
0		
0 1	✓	✓
⋮	⋮	⋮
0		

valid/Invalid $\begin{cases} 0 & \text{Invalid} \\ 1 & \text{valid} \end{cases}$

write back cache performance improvement:-

Dirty or modified	V/I	Tag	
☐	☐	☐	
☒	☒	☒	✓
⋮	⋮	⋮	⋮
☐	☐	☐	

Dirty { 0
1

if block is not modified

if block is dirty

write back the block from cache to mm, which have dirty bit 1.

$$\text{Tag directory size} = \text{no. of blocks in cache} * (\text{Tag} + \text{extra bits})$$

Ques) Direct mapping

mem add = 30 bits

cm size = 4K B

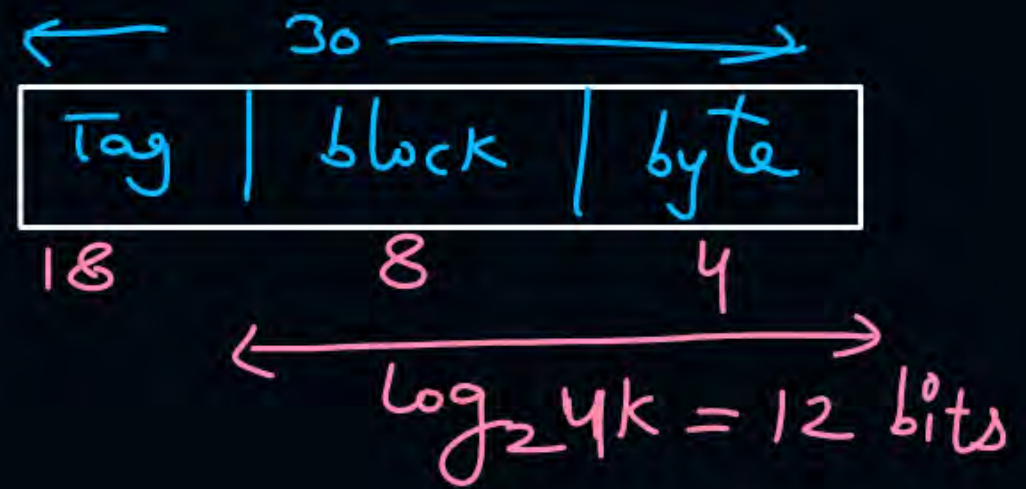
block size = 16 B = 2^4 B

for each block in cache tag, 1 valid/Invalid, 1 Modified bit stored.

Tag directory size = 5120 bits?

$$\hookrightarrow 2^8 * (18 + 1 + 1) \text{ bits}$$

$$= 256 * 20 \text{ bits} = \underline{\underline{5120 \text{ bits}}}$$





2 mins Summary



Topic

Cache Mapping

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Direct Mapping

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Tag

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Happy Learning

THANK - YOU